

Hybrid Physical Model and Status Data-Driven Dynamic Control for Digital Light Processing 3D Printing

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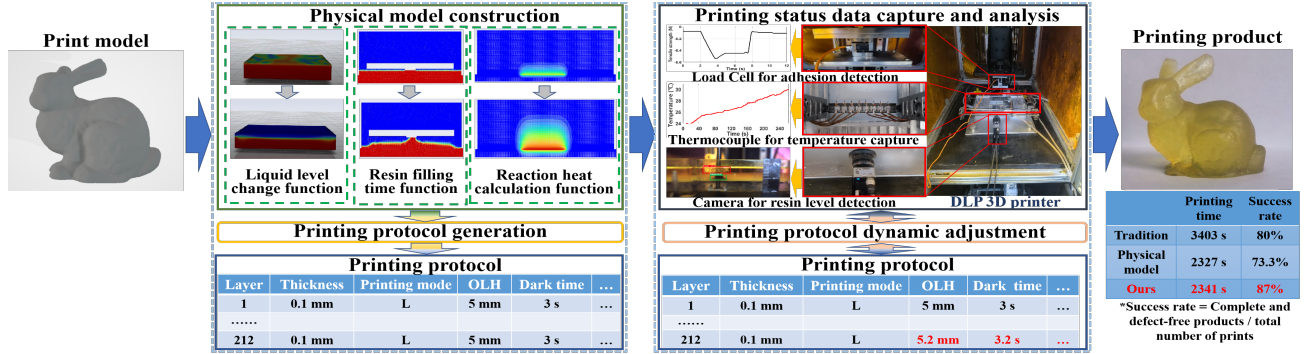


Figure 1: Overall schematic of the proposed method.

Abstract

In digital light processing (DLP) 3D printing, fixed control protocol parameters limit adaptability and lead to conservative settings, reducing printing efficiency. Existing methods that optimize parameters through physical models linked to the printing process do not account for real-time environmental changes, often resulting in potential print failures. This paper proposes a dynamic control scheme for DLP 3D printing that combines physical models with real-time data. Initially, a physical model of the printing process is developed to pre-generate the printing protocol. Then, a multimodal data capture scheme is designed to access the current state of the material and model. Finally, the printing protocol parameters are dynamically updated based on the analysis. Experimental results demonstrate that this method optimizes both printing time and success rate, significantly enhancing overall printing performance.

CCS Concepts

• Computing methodologies → Modeling and simulation.

Keywords

DLP 3D printing, physical model, status data, dynamic control

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1 Introduction

DLP 3D printing is popular for its speed and precision. However, current control protocols mainly rely on fixed, conservative parameters [Wu et al. 2018], limiting their adaptability and efficiency across models. Researchers have developed physical models to optimize protocols [Zhao et al. 2023], but static optimization may accumulate errors and cause failures, as it doesn't adapt to real-time conditions. Although sensors have been used for monitoring [Fang et al. 2022], they aren't integrated with protocols for proactive adjustments. To improve print quality and efficiency, it is essential to combine real-time data with dynamic optimization strategies.

This paper focuses on improving DLP 3D printing by monitoring key status data. We propose a dynamic control scheme that integrates physical models—covering resin filling, level state, and heat release—with real-time data. Mechanical, thermal, and optical sensors capture this data, which is then analyzed using data processing method. The control protocol is dynamically updated based on these insights. Refer to Fig. 1 for a visual overview.

2 METHODS

2.1 Physical model construction

The physical model includes the resin fill time calculation function (Eq. 1), the resin level change function (Eq. 2), and the heat release calculation function, which is based on Rehbein et al.'s autocatalytic photopolymerization model [Rehbein et al. 2020].

$$t_f(i) \approx H_s(t)/v_m + D_s(i) \cdot 12\mu_1 / H_s(t)^2 \rho g H_r(t) - H_s(t)^2 / 4v_m^2. \quad (1)$$

Eq. 1 is based on [Zhao et al. 2023]. μ_1 is resin's viscosity. $H_s(t)$ is the optimal lift height (OLH) for layer-wise printing, v_m is the platform speed, and $D_s(i)$ is the maximum fill distance (MFD) for slice i . Resin level H_r changes due to gravity are captured in Eq. 2:

$$H_r(t) = H_r(t - n) + \frac{K \times V_p + (1-K) \times (A_p \times d_p) + V_s - V_c}{A}, \quad (2)$$

where t is the total print time, n is one exposure cycle, and K is an indicator variable ($K = 1$ when the platform is fully immersed in resin). V_p is the platform volume, A_p is the platform's cross-sectional area, d_p is the depth of the platform in the resin, V_s is the cured model volume, and V_c is the solidified resin volume. A is the tank's cross-sectional area.

Additionally, we establish the curing time calculation function for different slices based on Jacobs' working curve in Eq. 3:

$$t_e(i) = \frac{E_c \times e^{C_d/D_p}}{I(i)}, \quad (3)$$

where E_c is the critical exposure, D_p is the penetration depth, and $I(i)$ is the homogenization light intensity [Wu et al. 2018].

2.2 Printing protocol generation

We extract four key control parameters from the printing protocol, all of which can be derived from the physical model. Table 1 outlines each parameter and its method of generation.

Table 1: DLP printing control parameters and their basis

Parameter	Basis for Generation
Printing mode	Use continuous printing if $D_s(i) < MFD$; otherwise, use layer-wise printing.
OLH	Jointly solve $D_s(i)$ with Eq. 1.
Exposure time	Solve based on layer thickness and Eq. 3.
Dark time	Account for resin fill time, platform movement time, and cooling time.

2.3 Printing status data capture and analysis

- **Adhesion force data:** A load cell is installed between the printing platform and the translation device captures the adhesion force. The Monte Carlo method calculates the cured layer's area to detect adhesion issues.
- **Resin temperature:** Eight K-type thermocouples on the translation device monitor temperature changes in and around the curing area. Positioned below the cured layer, they gather temperature data during dark time for layer-wise printing and correlate temperatures for continuous printing.
- **Resin level:** An optical camera with EPE and PA66 markers captures images of resin and fluorinated oil levels. For layer-wise printing, images are taken during dark time; for continuous printing, they are captured at fixed intervals. The YOLOv7 network detects the markers' positions to determine the resin level.

2.4 Printing protocol dynamic adjustment

Status data updates the physical model and adjusts parameters without interrupting printing. For layer-wise printing, updates occur during dark time. If adhesion issues arise, printing halts. In continuous printing, if temperature nears the monomer's flash point, printing pauses and fluorinated oil flow increases for cooling, resuming once heat dissipates.

3 RESULT AND DISCUSSION

To validate the impact of our method on printing efficiency, we printed 15 sets of rectangular and bunny models using the traditional method [Wu et al. 2018], a physical model optimized method [Zhao et al. 2023], and our proposed method. Table 2 records the average printing time and success rate. The first three layers used identical parameters. Our method, by accurately monitoring liquid level and other parameters, avoids print failures caused by parameter prediction errors (Fig. 2 (a) and (c) show underfilling defects caused by discrepancies between actual resin level changes and the physical model's predictions, leading to control parameters that fail to adapt to real printing conditions.). Our method improves print success rates by 29.1% on average for both model types, with a slight increase in printing time of 0.7%. Compared to the traditional method, it reduces printing time by 19.3% and increases success rates by 8.75%, enhancing printing efficiency.

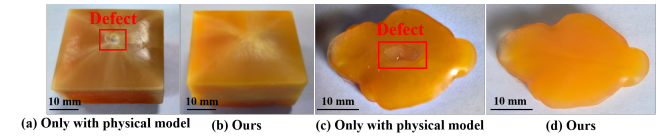


Figure 2: Comparison of surface morphology differences between different methods.

Table 2: Comparison of printing time and success rate with different methods

Model	Method	Printing time	Success rate
Rectangular	Traditional method	2115 s	87%
	Only with physical model	1818 s	66.7%
	Ours	1962 s	93%
Bunny	Traditional method	3403 s	80%
	Only with physical model	2327 s	73.3%
	Ours	2341 s	87%

4 CONCLUSION

This paper proposes a dynamic control scheme for DLP 3D printing that combines physical models with status data. By developing physical models, data capture and analysis methods, we dynamically adjust the printing protocol to improve efficiency.

Acknowledgments

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